ACADEMIC CITY UNIVERSITY COLLEGE
D5203: STATISTICAL METHODS FOR DATA SCIENCE

PREDICTING CUSTOMER CREDIT DEFAULT: A COMPARATIVE STATISTICAL STUDY OF PCA AND LOGISTIC REGRESSION IN COMMERCIAL BANKING

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1. Abstract

This study investigates the application of classical statistical methods in credit risk classification using the German Credit dataset. The research evaluates whether an interpretable framework combining Principal Component Analysis (PCA) and Logistic Regression can effectively manage high-dimensional financial data while maintaining regulatory transparency. Exploratory Data Analysis (EDA), normality testing (Shapiro-Wilk, Q-Q plots, and P-P plots), and log transformations were conducted to prepare the dataset. PCA reduced 61 one-hot encoded variables to 32 principal components while retaining 95% of the total variance. A Logistic Regression classifier achieved a test accuracy of 77% and an AUC-ROC of 0.80. Confusion matrix analysis revealed that Recall remains the most critical metric due to the high financial cost of undetected defaults. The findings demonstrate that classical statistical techniques remain robust, interpretable, and regulator-aligned tools for credit risk modeling.
The study demonstrates that interpretable statistical methods remain viable alternatives to complex black-box models in regulated financial environments.

2. Background

The evolution of credit risk assessment represents one of the most important breakthroughs in applied statistical decision-making. Historically, lending decisions were based largely on subjective judgment, personal relationships, and limited financial records. This approach often resulted in inconsistent and biased credit allocation.

The development of formal credit scoring systems, notably those pioneered by the Fair Isaac Corporation, marked a fundamental shift toward probabilistic modeling. Later, international regulatory frameworks introduced by the Basel Committee on Banking Supervision (Basel Committee on Banking Supervision, 2011) institutionalized the use of quantitative risk assessment in banking. Under Basel II and Basel III, financial institutions are required to estimate Probability of Default (PD) (Basel Committee on Banking Supervision, 2011), making statistical modeling central to regulatory compliance.

In recent years, machine learning has introduced complex algorithms such as random forests and neural networks into financial modeling. While these models sometimes improve predictive accuracy, they often lack interpretability. In regulated industries such as banking, interpretability is not optional—it is a requirement. Institutions must justify lending decisions to regulators and customers.

This creates a methodological tension: should financial institutions prioritize marginal improvements in predictive performance using opaque models, or should they adopt interpretable statistical frameworks that align with regulatory expectations?

This study investigates whether a classical approach—Principal Component Analysis (PCA) combined with Logistic Regression—can provide strong predictive performance while maintaining transparency and statistical rigor. By applying this framework to the German Credit

dataset, the study evaluates both predictive effectiveness and financial implications of classification errors.

3. Problem Statement

Despite the growing dominance of black-box machine learning models in financial analytics, there remains a significant gap in balancing predictive performance with interpretability in high-dimensional credit datasets.

Financial datasets frequently contain numerous categorical variables that require one-hot encoding, leading to dimensional expansion. High dimensionality introduces multicollinearity, unstable coefficient estimation, and potential overfitting. Many studies focus primarily on maximizing classification accuracy without adequately addressing:

- The structural stability of interpretable models in high-dimensional settings,
- The role of dimensionality reduction before classification,
- The asymmetric financial cost of misclassification errors.

While complex ensemble models may provide slight improvements in predictive metrics, they often sacrifice transparency. There is limited applied research demonstrating how dimensionality reduction techniques such as PCA can enhance classical Logistic Regression performance while maintaining interpretability and regulatory alignment.

This study addresses this gap by implementing PCA prior to Logistic Regression and evaluating performance using both statistical and financial perspectives.

4. Objectives

4.1 General Objective

To develop and evaluate an interpretable statistical framework for predicting credit default risk using PCA and Logistic Regression.

4.2 Specific Objectives

1. Conduct exploratory data analysis to understand feature distributions and relationships.
2. Test for normality using P-P plots, Q-Q plots, and the Shapiro-Wilk test.
3. Apply log transformations to address skewness in numerical variables.
4. Implement PCA to reduce dimensionality after one-hot encoding.
5. Train a Logistic Regression classifier.
6. Evaluate model performance using accuracy, ROC-AUC, precision, recall, and F1-score.
7. Interpret results within a financial risk management context.

5. Significance of the Study

This study is significant for three main reasons:

Methodological Significance

It demonstrates how classical statistical techniques remain competitive when applied systematically and diagnostically.

Practical Significance

The model provides a transparent framework suitable for regulatory environments where decision traceability is essential.

Policy Significance

By emphasizing recall and cost-sensitive evaluation, the study aligns predictive modeling with financial stability objectives and systemic risk mitigation.

6.1 Theoretical Review

Information Asymmetry and Adverse Selection

The foundation of credit risk modeling lies in Information Asymmetry Theory. Akerlof (1970), in his seminal paper *"The Market for Lemons,"* demonstrated how markets collapse when one party possesses superior information. In credit markets, borrowers have more accurate knowledge of their financial stability and repayment capacity than lenders.

This imbalance produces adverse selection, where high-risk borrowers are more likely to seek loans, while lenders cannot perfectly distinguish between safe and risky applicants. Without structured screening mechanisms, financial institutions may misprice loans or accumulate excessive default risk.

Credit scoring models serve as screening devices that transform observable borrower characteristics into measurable default probabilities. In doing so, they reduce informational inefficiency and improve allocation of credit.

Thus, the theoretical justification for this study lies in reducing information asymmetry through statistical classification.

Logistic Regression as a Probability Model of Default

Logistic regression provides a probabilistic framework consistent with financial risk modeling. This formulation aligns directly with the concept of Probability of Default (PD) used in regulatory capital frameworks (Basel Committee on Banking Supervision, 2011).

Unlike purely predictive algorithms, logistic regression produces interpretable coefficients, allowing financial institutions to understand the directional effect of each risk factor. Interpretability is particularly important in regulated banking environments where model transparency is required.

Dimensionality Reduction and Multicollinearity

Financial datasets often contain correlated variables such as income, loan size, and debt ratios. Multicollinearity can inflate coefficient variance and destabilize maximum likelihood estimation.

Principal Component Analysis (PCA) addresses this issue by transforming correlated predictors into orthogonal components that maximize variance retention (James et al., 2013). By constructing linear combinations that capture the greatest variance in the data, the transformation improves numerical stability while preserving essential information.

In this study, PCA serves as a theoretical enhancement to logistic regression by strengthening its statistical reliability.

6.2 Empirical Review

Benchmark Credit Datasets in Risk Modeling

The German Credit dataset from the UCI Machine Learning Repository has been widely used as a benchmark for evaluating credit scoring algorithms. Numerous empirical studies test classification models on structured credit datasets like the one used in this study.

Research consistently shows that logistic regression performs competitively on tabular financial datasets, especially when features are properly engineered and scaled.

Statistical Models vs Machine Learning

Recent empirical studies compare traditional statistical models with machine learning algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting.

Findings generally indicate:

- Machine learning models may achieve slightly higher accuracy or AUC.

- However, gains are often marginal on moderate-sized structured datasets.
- Interpretability decreases substantially with model complexity.
- Regulatory acceptance favors transparent statistical models.

In banking environments, the trade-off between interpretability and predictive power is critical. As a result, logistic regression remains widely implemented in credit risk modeling despite advancements in machine learning.

Identified Research Gap

Although prior research compares logistic regression and machine learning, limited applied studies focus on:

- Integrating PCA with logistic regression to enhance stability
- Evaluating threshold adjustments from a financial cost perspective
- Aligning classification metrics with credit loss implications

This study contributes by:

1. Combining PCA with logistic regression.
2. Conducting threshold experiments to balance precision and recall.
3. Interpreting classification errors in financial terms rather than purely statistical metrics.

6.3 Summary of Literature

The literature establishes three key foundations:

1. Information asymmetry necessitates structured credit screening.
2. Logistic regression provides a theoretically consistent probabilistic model of default.
3. While machine learning models offer performance improvements, interpretability remains critical in financial risk management.

However, limited applied research integrates dimensionality reduction with cost-sensitive evaluation in credit classification.

This study addresses that gap by developing a PCA-enhanced logistic regression framework and evaluating its performance using both statistical and financial perspectives.

7. Methodology

7.1 Data Acquisition

This study utilizes a version of the **German Credit dataset**, obtained from the UCI Machine Learning Repository (Hofmann, 1994). The dataset contains **1,000 loan applicants**, each classified as either:

- 0 = Good credit risk
- 1 = Bad credit risk

Each row represents a single applicant, described by demographic, financial, and credit-history attributes, together with a credit-risk label typically used in retail credit scoring models.

The dataset consists of:

- 7 numerical variables
- 14 categorical variables

The available features are written below:

- **status_account**: Status of existing checking account in DM(Deutschen Mark)
- **month_duration**: Duration in months for the loan/credit
- **credit_history**: credit history of the applicant
- **purpose**: purpose of the credit
- **credit_amount**: credit amount of the loan
- **status_savings**: status of savings account in DM(Deutschen Mark)
- **years_employment**: Present employment since
- **payment_to_income_ratio**: Installment rate in percentage of disposable income
- **status_and_sex**: categories with personal status and sex
- **secondary_obligor**: Other debtors / guarantors
- **residence_since**: Present residence since
- **collateral**: possible collateral for loan
- **age**: age in years
- **other_installment_plans**: Other installment plans
- **housing**: indicator of the current housing (rent, own or for free)
- **n_credits**: Number of existing credits at this bank
- **job**: categories of job
- **n_guarantors**: Number of people being liable to provide maintenance for
- **telephone**: flag to indicate if the customer have a telephone registered in the customer name
- **is_foreign_worker**: flag indicating foreign worksers
- **target**: good for customers that properly paid the loan, bad otherwise

After preprocessing and one-hot encoding of categorical features, the feature space expanded to **61 predictors**.

The German Credit dataset is widely used in benchmarking credit scoring models, making it appropriate for evaluating classification performance and methodological enhancements such as PCA.

The dataset is designed to support:

- Binary credit-risk classification (good vs. bad credit)

- Evaluation of credit-scoring algorithms
- Research in bias, fairness, and explainability in financial decision systems

7.2 Study Variables

Dependent Variable

The dependent variable is binary:

$$Y = \begin{cases} 1 & Bad\ (High\ Risk) \\ 0 & Good\ (Low\ Risk) \end{cases}$$

This aligns with binary classification modeling using logistic regression.

Independent Variables

The independent variables represent borrower characteristics including:

- Credit amount
- Loan duration
- Age
- Employment status
- Housing status
- Existing credit history
- Savings account status
- Personal status and other socio-financial indicators

Categorical variables were transformed using one-hot encoding, increasing dimensionality but preserving information content.

7.3 Data Cleaning and Preprocessing

Missing Value Assessment

An initial audit confirmed that **no missing values** were present in the dataset. Therefore, imputation was not required.

Skewness and Log Transformation

Exploratory Data Analysis revealed that key numerical variables were positively skewed. Particularly **Credit Amount, Duration,** and **Age.**

To stabilize variance and reduce right skewness, log transformation was applied

This transformation:

- Reduced the influence of extreme values
- Improved approximation to normality
- Enhanced model stability

7.4 Normality Testing

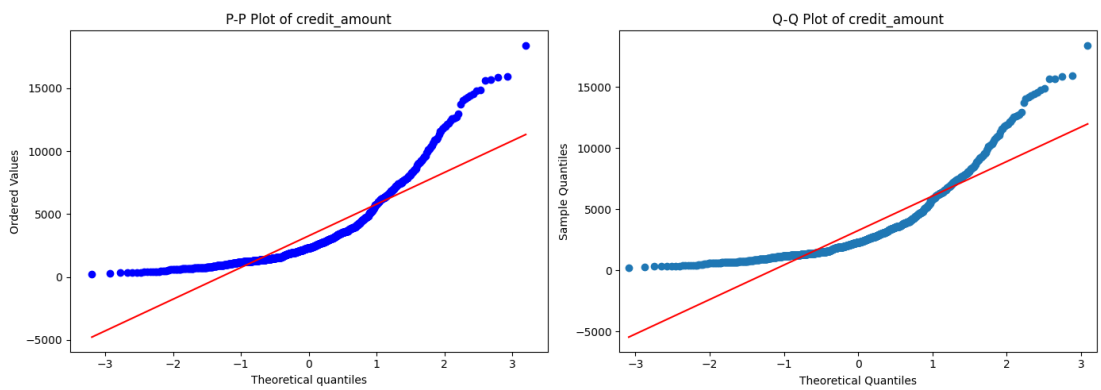
To formally evaluate distributional properties, the **Shapiro–Wilk test** was conducted on numerical variables.

Results showed:

- p-values < 0.05 for key variables
- Indicating statistically significant deviation from normality

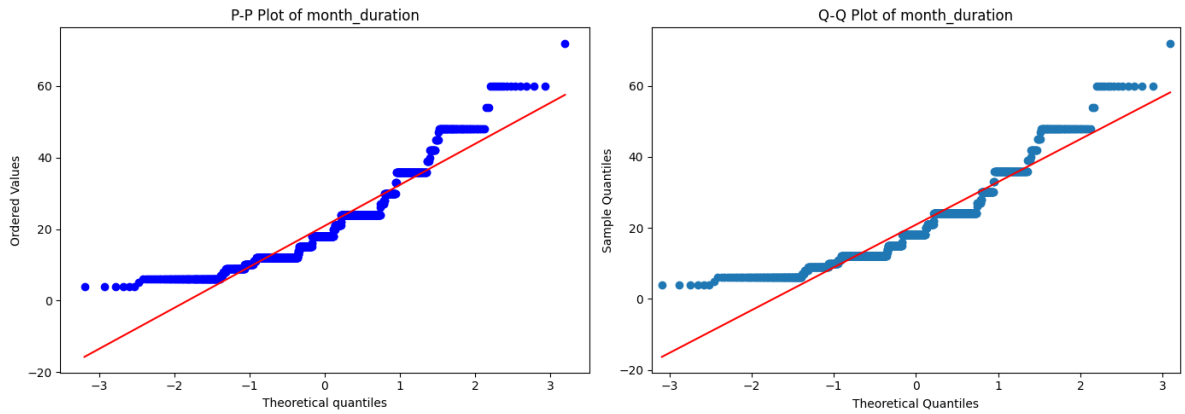
Q–Q plots and P–P plots further confirmed right-skewed distributions.

Q-Q and P-P plots for Credit Amount plots:



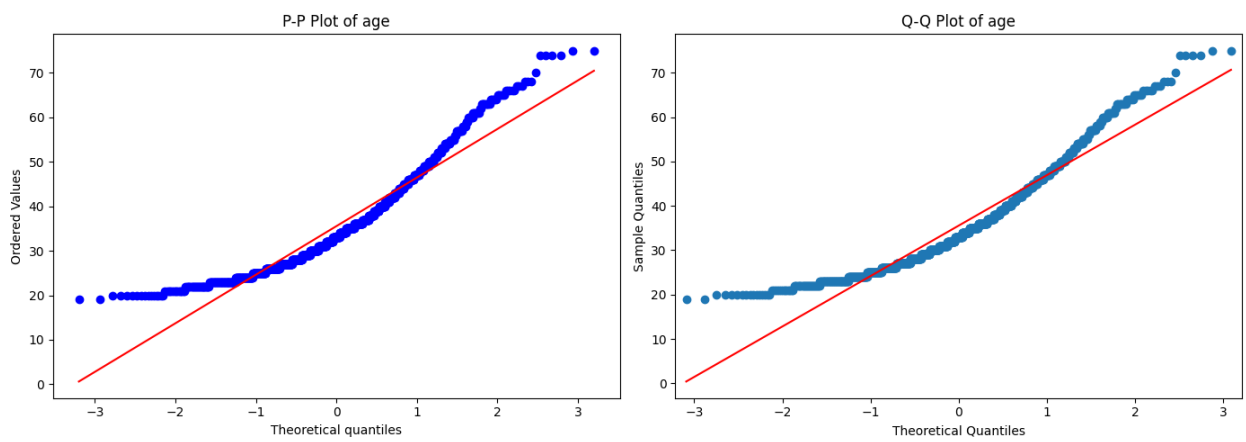
Shapiro-Wilk Test Statistic for credit_amount: 0.7934
Shapiro-Wilk P-value for credit_amount: 0.0000
Conclusion: **credit_amount** is likely not normally distributed ($p < 0.05$)

Q–Q and P–P plots for Month duration:



Shapiro-Wilk Test Statistic for month_duration: 0.8998
Shapiro-Wilk P-value for month_duration: 0.0000
Conclusion: **month_duration** is likely not normally distributed ($p < 0.05$)

Q–Q and P–P plots for Age:



Shapiro-Wilk Test Statistic for age: 0.9175
Shapiro-Wilk P-value for age: 0.0000
Conclusion: **Age** is likely not normally distributed ($p < 0.05$)

7.5 Feature Scaling

All numerical features were standardized using **StandardScaler**. Log-transformation (log1p) was applied to skewed features. Features were standardized using.

$$z = (x - \mu)/\sigma$$

Standardization was necessary because:

- PCA is variance-sensitive
- Logistic regression coefficients become more stable

- Features contribute equally during model estimation

Scaling parameters were derived from the training set only to prevent data leakage.

7.6 Dimensionality Reduction: Principal Component Analysis (PCA)

Following encoding and scaling, the dataset contained **61 predictors**, creating potential multicollinearity.

PCA was applied to:

- Reduce dimensionality
- Remove correlated redundancy
- Improve computational efficiency
- Enhance logistic regression stability

Components were retained based on cumulative explained variance.

The analysis showed that **95% of total variance** was retained using **32 principal components**, effectively reducing dimensionality by nearly 50%.
This transformation preserved informational content while minimizing overfitting risk.

7.7Classification Model: Logistic Regression

The processed dataset was split into 80% Training set and 20% Testing set.

Logistic regression was implemented as the classification model. Parameters were estimated using Maximum Likelihood Estimation (MLE).

Logistic regression was selected due to:

- Interpretability
- Probabilistic output (aligned with PD modeling)
- Regulatory acceptance in financial institutions
- Competitive empirical performance on structured datasets

7.8 Model Evaluation and Diagnostic Testing

Model performance was evaluated using:

- Accuracy
- Precision
- Recall

- F1-score
- ROC-AUC
- Confusion Matrix

Confusion Matrix Interpretation

The confusion matrix distinguishes:

- True Positives (correctly identified bad borrowers)
- False Negatives (missed high-risk borrowers)
- False Positives (good borrowers incorrectly rejected)
- True Negatives (correctly approved borrowers)

	Predicted Good	**Predicted Bad**
Actual Good	123 (TN)	18 (FP)
Actual Bad	28 (FN)	31 (TP)

Threshold Experimentation

Because false negatives in credit risk represent potential financial loss, threshold sensitivity analysis was conducted to balance:

- Recall (risk detection strength)
- Precision (approval efficiency)

This experiment aligns predictive performance with real-world financial trade-offs.

8. Results and Discussions

8.1 Exploratory Data Analysis

The exploratory data analysis revealed important structural characteristics of the dataset. The average credit amount was 3,271 Deutsche Marks (DM), with loan durations averaging approximately 20 months and an average borrower age of 35 years.

No missing values were detected across variables, indicating strong data completeness and eliminating the need for imputation techniques. However, distributional analysis revealed pronounced right-skewness in financial variables such as **credit amount** and **loan duration.**

This skewness is economically intuitive. Most borrowers request relatively small loans, while a minority demand significantly larger amounts. From a risk perspective, large loan amounts introduce heavier exposure, making distributional characteristics particularly important in predictive modeling.

The class distribution also reflects mild imbalance, with **'Good'** credit risks forming the majority. This imbalance has implications for accuracy interpretation, as naive classifiers may achieve artificially inflated accuracy by favoring the majority class.

Thus, EDA provided both statistical and economic insight into the structure of credit demand and risk exposure.

8.2 Normality Testing: Shapiro -Wilk and Q-Q and P-PP plot

To formally evaluate distributional assumptions, Shapiro-Wilk tests were applied to **age**, **credit amount**, and **loan duration**. In each case, p-values were less than **0.05**, leading to **rejection of the null hypothesis** of normality.

Q-Q and P-P plots further confirmed deviation from normality, particularly in the upper tails, indicating heavy right-skewness.

Although Logistic Regression does not require normality of predictors, severe skewness can influence coefficient stability and optimization convergence. Therefore, log-transformation (log1p) was applied to skewed variables.

8.3 PCA Variance Retention and Component Selection

One-hot encoding expanded categorical predictors, increasing dimensionality to 61 features. High dimensionality introduces multicollinearity and reduces the efficiency of maximum likelihood estimation in Logistic Regression.

Principal Component Analysis was therefore implemented to:

1. Orthogonalize predictors,
2. Reduce redundancy,
3. Improve numerical stability.

The cumulative explained variance plot indicated that **32 principal components** were required to retain 95% of total variance.

This result demonstrates that nearly half of the original encoded dimensions contained redundant information. By reducing **61 predictors to 32 orthogonal components**, the model achieves computational efficiency without substantial information loss.

However, this transformation introduces a trade-off: while statistical stability improves, direct interpretability of individual financial characteristics is reduced because components are linear combinations of original variables.

This trade-off reflects a central tension in statistical modeling between interpretability and efficiency.

8.4 Model Performance Evaluation

8.4.1 Confusion Matrix and Classification Metrics

On the test dataset, the Logistic Regression model achieved:

- Accuracy: 77%
- ROC-AUC: 0.80
- Precision: 0.63
- Recall: 0.53
- F1-score: 0.58

The confusion matrix is summarized below:

	Predicted Good	**Predicted Bad**
Actual Good	123 (TN)	18 (FP)
Actual Bad	28 (FN)	31 (TP)

- True Negatives (Correctly identified Good borrowers): 123
- True Positives (Correctly identified Bad borrowers): 31
- False Positives (Incorrectly rejected Good borrowers): 18
- False Negatives (Missed defaulters): 28

An **accuracy** of 77% indicates overall strong predictive performance. However, accuracy alone is insufficient in risk-sensitive domains because it assumes equal cost of errors.

Precision of 0.63 indicates that when the model flags a borrower as high-risk, it is correct 63% of the time.

Recall of 0.53 indicates that only 53% of actual defaulters are captured. This suggests that nearly half of high-risk borrowers remain undetected at the default threshold of 0.50. In financial contexts, this represents material credit exposure.

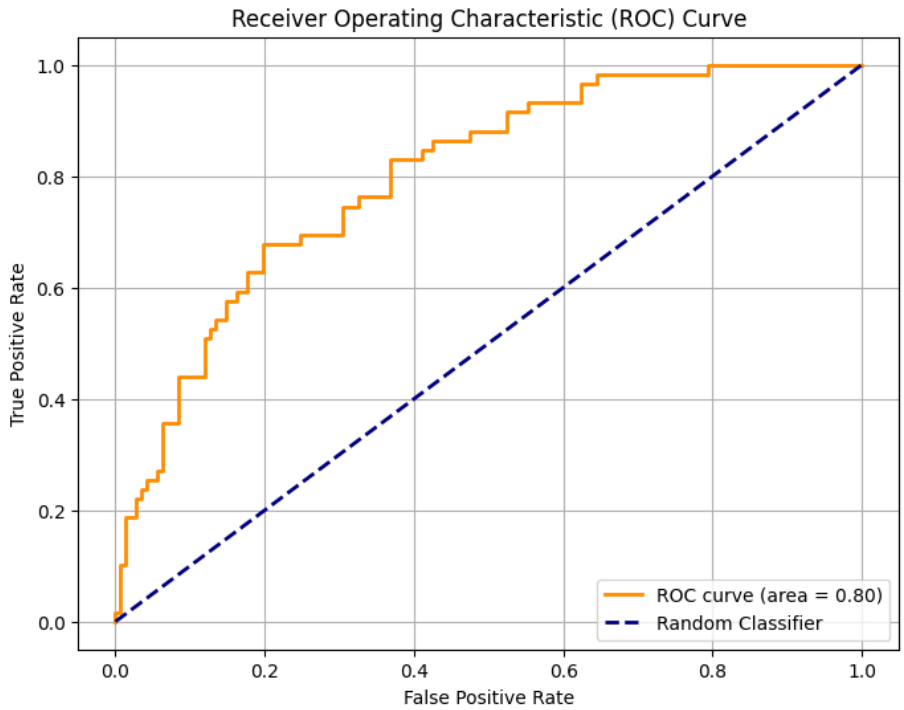
The Diagnosis Analogy – Most Important Metric

As stated in the notebook, identifying a "Bad" credit risk is similar to a **medical diagnosis**. A **False Negative (Type II Error)**—approving a borrower who then defaults—is significantly more costly than a **False Positive**. The bank loses the entire loan principal in a default, whereas a false rejection results in a lost interest opportunity.

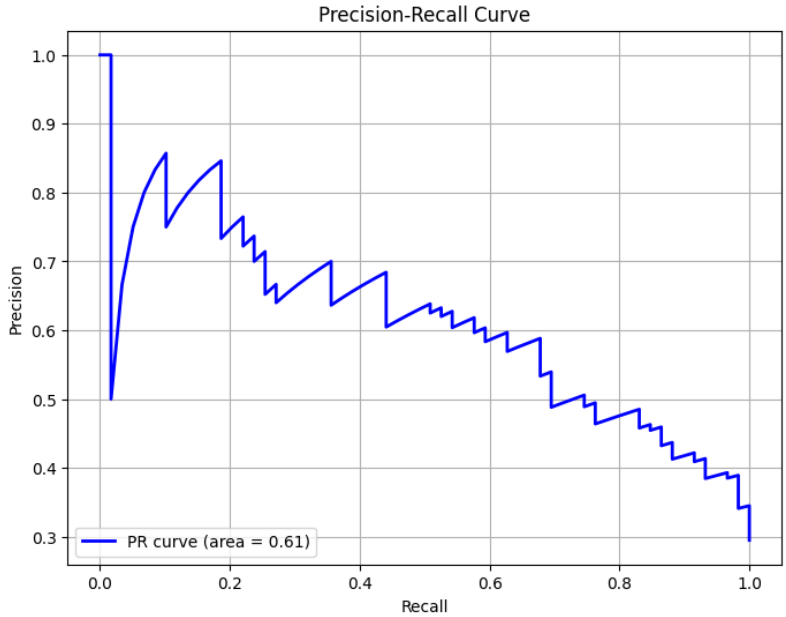
Therefore, **Recall** becomes the most important and critical metric because False Negatives (missed defaulters) expose the bank to substantial financial loss.

8.4.2 ROC and Precision-Recall Curve Interpretation

The **ROC-AUC value of 0.80** indicates strong discriminatory power. This means that the model is effective at ranking borrowers by risk even if the final classification threshold affects binary decisions.



The **Precision-Recall curve (AUC-PR = 0.61)** reflects the challenge of identifying the minority high-risk class. In moderately imbalanced datasets, PR curves provide more insight into minority class detection performance.



The divergence between ROC and PR performance highlights the difficulty of balancing false alarms with missed defaults.

8.4.3 Threshold Experiment: Balancing Precision and Recall

To further explore classification trade-offs, the decision threshold was reduced from **0.50** to **0.35**.

This adjustment increased **Recall** from **0.53** to **0.69**, capturing 10 additional defaulters. However, False Positives increased significantly, reducing overall accuracy to **69%**.

This experiment demonstrates the fundamental **trade-off between sensitivity and precision**. Lowering the threshold widens the detection net but increases false alarms.

From a financial decision-making perspective, the optimal threshold should not maximize accuracy but minimize expected cost:

$$\boldsymbol{Expected\ Cost} = (FN\ x\ Cost_{FN}) + (FP\ x\ Cost_{FP})$$

If the financial cost of default substantially exceeds the profit from a performing loan, prioritizing higher recall becomes economically rational.

Thus, threshold tuning transforms the model from a purely statistical tool into a financial decision instrument.

Performance Comparison

Metric	Threshold 0.50	Threshold 0.35	Change
True Positives (TP)	31	41	+10
False Negatives (FN)	28	18	−10
False Positives (FP)	18	43	+25
Recall (Class 1)	0.53	0.69	+0.16
Overall Accuracy	0.77	0.69	−0.08

9. Summary, Conclusion and Recommendations

9.1 Summary of Findings

This study examined credit default risk prediction using the German Credit dataset, applying Principal Component Analysis (PCA) in combination with Logistic Regression to classify borrowers as Good (low risk) or Bad (high risk).

The study was grounded in Information Asymmetry Theory, which explains the need for structured screening mechanisms in lending markets. Logistic regression was selected due to its probabilistic interpretation and regulatory relevance, while PCA was introduced to reduce dimensionality and mitigate multicollinearity arising from one-hot encoded categorical variables.

Key methodological steps included:

- Log transformation of skewed financial variables
- Standardization of features
- Dimensionality reduction from 61 predictors to 32 principal components while retaining 95% of total variance
- Model evaluation using Accuracy, Precision, Recall, F1-score, ROC-AUC, and threshold experimentation

The results indicate that the PCA-enhanced logistic regression model demonstrates strong discriminatory ability. However, performance analysis revealed an important trade-off between precision and recall — particularly in detecting high-risk borrowers.

The confusion matrix highlighted that false negatives (high-risk borrowers misclassified as low risk) represent potential credit loss exposure, whereas false positives represent opportunity cost through rejection of creditworthy applicants.

Threshold experimentation showed that predictive performance is not fixed; rather, it can be adjusted depending on institutional risk appetite.

9.2 Conclusion

This study confirms that logistic regression remains a robust and effective model for structured credit risk classification, particularly when enhanced through dimensionality reduction techniques such as PCA.

The findings support three major conclusions:

1. Dimensionality reduction improves numerical stability without significantly sacrificing predictive performance.
2. Logistic regression provides interpretable probability estimates aligned with financial risk management frameworks.
3. Model evaluation must extend beyond overall accuracy and incorporate financial cost considerations of classification errors.

Although more complex machine learning algorithms may yield marginal accuracy improvements, the interpretability, transparency, and regulatory alignment of logistic regression make it highly suitable for retail credit scoring environments.

Ultimately, predictive modeling in credit risk is not purely a statistical exercise — it is a financial decision-making tool that must balance default detection with business growth objectives.

9.3 Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. Consider Cost-Sensitive Threshold Selection

Financial institutions should not rely solely on the default 0.5 classification threshold. Instead, thresholds should be adjusted based on:

- Expected loss from false negatives
- Opportunity cost of false positives
- Institutional risk tolerance

A lower threshold may improve detection of high-risk borrowers at the expense of rejecting more good applicants.

2. Integrate Interpretability into Risk Governance

Given regulatory requirements and risk governance standards, institutions should prioritize models that provide explainable outputs. Logistic regression remains appropriate where transparency is critical.

3. Combine Statistical Rigor with Financial Strategy

Future implementations should incorporate:

- Expected loss modeling
- Profit-based evaluation metrics
- Sensitivity analysis under different economic scenarios

This would further align predictive performance with financial decision-making.

4. Future Research Direction

Future studies may explore:

- Comparison with advanced machine learning models
- Incorporation of macroeconomic variables
- Use of cross-validation techniques for enhanced robustness
- Cost-sensitive learning algorithms

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